Capstone Project

Technology Core Java – L2

Bank Application

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1. Project Overview

Create console-based Bank Application, which helps the bank relationship managers to create customer profile, manage customer money transfer.

2. Project Detail

Create **console-based application** to perform the below functions

- Create customer
- Bank Transfer
- Report

3. Project Menu

Main Menu consists of below options to choose

- Create Customer
- Bank Transfer
- Report
- Exit

Please note: All Submenu should have option to go **Back to Main Menu**

3.1. Create Customer

Enter Customer Details to insert (**Comma** separated fields as given in table, one row for each customer, enter one customer at a time then go back to Main Menu)

S.No	Field	Detail
1	Customer Name	100 Char String
2	Date of Birth	DD-MM-YYYY

3	Email	100 Char,
4	Phone Number	Number
5	Account Type	Current, Savings, or DMAT
6	Account Status	Active, Inactive, or Suspended
7	Balance amount	<10 Digits

3.2. Bank Transfer

Transfer fund from one account to another. Provide below 3 fields as **comma** separated.
<From ACCOUNT_NO>,<To ACCOUNT_NO>, <AMOUNT>

3.3. Report

- Report (Get below details with pagination of 5 rows at a time)
- Entering space/enter/(any custom key configuration) should retrieve next set of 5 rows and displays.
- Bank relationship manager could use preconfigured key to exit from the system.

S.No	**Customer Name**	**Account Number**	Balance	Last Updated Date/Time	**Last Transaction**
1	< USTOMER_NAME>	<ACCOUNT_NO>	< BALANCE>	<UPDATED_TIMESTAMP> Formatted in below form 31 May 2024 2:30 PM	Credited/Withdrawn

4. Project Table Details

Project will utilize below tables, if you want to add more table/constraints to support more function or features, please do for your learning.

4.1. Table: CUSTOMER

Column	Data Type	Length	Description
ID	INT	10	A **unique** ID that represents the customer of the bank. **Auto generated number, increment by 1001** This can be a **primary key** and used to access other tables.
NAME	VARCHAR	100	Name of the customer.
DOB	DATE		Date of birth of the customer.
EMAIL	VARCHAR	100	Email ID of the customer
PHONE	VARCHAR	15	Phone number of the customer

4.2. Table: ACCOUNT

Column	Data Type	Length	Description
ACCOUNT_NO	BIGINT	17	Account number, which is a **unique** ID, to identify the account of the customer. Create through Java Date TimeStamp in below format **MMddHHmmssSSSSSSS**
ACCOUNT_TYPE	VARCHAR	10	The type of the account. For example, Current, Savings, or DMAT.
CUSTOMER_ID	INT	10	Used as a **referential key** to associate a customer to the account. (Refer to CUSTOMER Table)
ACCOUNT_STATUS	VARCHAR	10	Status of the account. For example, Active, Inactive, or Suspended.
UPDATED_TIMESTAMP	DATETIME		The date when the account was active for use.
BALANCE	INT	10	The total available balance in the account.

4.3. Table: TRANSACTION_DETAILS

Column	Data Type	Length	Description
ACCOUNT_NO	BIGINT	17	Account number, which is a unique ID, to identify the account of the customer.
CUSTOMER_ID	INT	10	Used as a **referential key** to associate a customer to the account. (Refer to CUSTOMER Table)
AMOUNT	INT	10	The amount withdrawn at a time.
TRANSACTION TYPE	CHAR	10	CREDIT/DEBIT
UPDATED_TIMESTAMP	DATETIME		The time when the amount was withdrawn.

5. Project Guidelines

The features given below are the **minimum basic features** that are required to be implemented. But you need not limit yourself for your exploration.

- Validations must be done, wherever applicable, using separate custom Util class method.
- Provide custom user-friendly error messages on error, should not show default java error/stack
- Create reusable methods, follow best practices in naming variable, method, etc
- Create as executable jar
- Create appropriate Util classes for use DateUtil, StringUtil, FileUtil, etc
- Ensure to fix coding standard issues if any, leverage tools like SonarQube
- Ensure to follow the given file name convention for all CSV files
- Avoid duplicating the code, instead cross refer using best practice
- Use Java Version 8/latest & supported JDBC & MySQL
- Use Prepared Statement
- Ensure Table has required constraints & following best practice

6. General Guidelines

- Any tools & software use Wipro WASP/IMG Request instead of downloading from Internet on Wipro Asset.

7. Capstone Project Submission

- After the implementation and testing of your Project is complete, you need to upload the project along with the screen shots of the output & files onto the given location (MS Team / Wipro One Drive).
- Rename the project name to end with you employee number (ex. BankApplication-123456).
- You also need to fill the form providing the following details:
  - Employee Number (After filling, check if the employee number is correct)
  - Your Wipro Email ID (Not AD User Name)

8. References

- SonarQube: https://www.sonarsource.com/products/sonarqube/
- Coding Guidelines
  - https://www.oracle.com/java/technologies/javase/codeconventions-programmingpractices.html
  - https://www.geeksforgeeks.org/coding-guidelines-in-java/
  - https://www.javatpoint.com/coding-guidelines-in-java
  - https://www.cs.cornell.edu/courses/JavaAndDS/JavaStyle.html
  - https://dzone.com/refcardz/jdbc-best-practices
  - https://docs.oracle.com/en/database/oracle/oracle-database/19/tdpjd/best-practices.html